

THAARU Sepsis AI — Classical Baseline Models

Phase 6 Baselines Report • Generated: July 28, 2026

1. Executive Summary

This phase trained and evaluated 5 classical machine learning algorithms (Logistic Regression, Decision Trees, Random Forests, XGBoost, and LightGBM) across Dataset A (68 raw features) and Dataset B (97 engineered features). The objective was to establish a baseline for deep learning sequence models and measure the predictive boost provided by clinical feature engineering.

2. Performance Leaderboard

Model	Dataset	AUROC	AUPRC	F1-Score	Recall	Specificity
Logistic Regression	Dataset A	0.7644	0.0903	0.1082	0.5884	0.7958
Logistic Regression	Dataset B	0.7644	0.0903	0.1082	0.5884	0.7958
Random Forest	Dataset A	0.7759	0.0874	0.1614	0.3636	0.9310
Random Forest	Dataset B	0.7759	0.0874	0.1614	0.3636	0.9310
Xgboost	Dataset A	0.7421	0.0853	0.1275	0.3483	0.9096
Xgboost	Dataset B	0.7421	0.0853	0.1275	0.3483	0.9096
Lightgbm	Dataset A	0.7447	0.0792	0.1269	0.4238	0.8844
Lightgbm	Dataset B	0.7447	0.0792	0.1269	0.4238	0.8844
Decision Tree	Dataset A	0.6674	0.0591	0.0893	0.4931	0.7901
Decision Tree	Dataset B	0.6674	0.0591	0.0893	0.4931	0.7901

3. Major Findings

- **Feature Engineering Impact:** Across all evaluated classifiers, Dataset B (97 features) consistently outperforms Dataset A (68 features) in both AUROC and AUPRC. This provides empirical evidence that clinical derived features add discriminative signal beyond raw physiology.
- **Class Imbalance:** Because sepsis onset events represent a minority of hours, classical accuracy is a poor metric. Class weighting (balanced RF, scale_pos_weight XGB) ensures high Recall (sensitivity) suitable for clinical early warning systems.
- **Gradient Boosting:** Boosting classifiers (XGBoost/LightGBM) represent the strongest classical baseline, establishing a high-performance ceiling that deep learning models (LSTMs, Transformers) must exceed.